

SEN 4015: Advanced Python Programming Term Project

Formal Guidelines and Requirements

Phase 1: Research and Topic Discovery

The initial phase focuses on academic and technical grounding. You are required to transition from general Python concepts to a specific, research-backed project objective.

- **Literature Review:** You will work in groups of 2 to 4 members to implement a project addressing a real-world problem using Python. You must select and critically analyze at least two technical sources. These should be either peer-reviewed scientific papers (e.g., from IEEE, ACM, or arXiv) or significant open-source repositories (e.g., specialized libraries on GitHub).
- **Topic Identification:** The objective is to identify a specific problem or gap that can be solved with Python. This could include domains such as **Machine Learning (Scikit-learn)**, **Data Analysis (Pandas/NumPy)**, or **Advanced Scripting (Multithreading/RegEx)**. Your project must be a direct extension, implementation, or improvement of the concepts found in your selected literature.

Phase 2: Project Development and Formal Documentation

The development phase culminates in the submission of your final Python implementation and a comprehensive technical report.

- **Final Report Structure:**
 - **Foundational Analysis:** A deep dive into the paper or repository that served as the basis for your project.
 - **Core Problem:** What specific challenge or data question is being addressed?
 - **Technical Approach:** What Python methodologies, algorithms, or libraries were used?
 - **Key Findings:** What was the significance of the original work?
 - **Technical Implementation:** An exhaustive explanation of your software's architecture, including your use of Object-Oriented Programming (Classes, Inheritance) and core logic.
 - **Innovation & Contribution:** A dedicated "Innovation" section detailing your unique additions, optimizations, or enhancements to the existing work.
- **Submission Protocol:**
 - Exactly one member from each group is responsible for the official submission.

- **Deadline:** The end of the last third week of the semester.
- **Format:** The report (PDF) and the source code (.py files) must be uploaded as separate, individual files.
- **Restriction:** Compressed formats (e.g., .zip, .rar) are strictly prohibited and will not be accepted.

Phase 3: Group Registration and Scheduling

Administrative compliance is mandatory for participation in the final phase.

- **Teams Registration:** A Google Sheet will be provided on the Teams General channel.
- **Procedure:** A single representative must record the project title and the full names of all members.
- **Eligibility:** Completion of this sheet is a prerequisite for the presentation. Groups that do not register by the deadline will be disqualified.
- **Finalized Schedule:** I will assign each group a slot during the final two weeks of class (W13–W14). Once published, the schedule is final.

Phase 4: Oral Defense and Presentation

The final two weeks of the semester are reserved for the oral defense.

- **Mandatory Attendance:** Every group member must be present and actively participate.
- **Performance:** All members must be fully conversant with both the theoretical background and the practical code implementation.
- **Presentation Slides:** Groups must prepare professional slides covering the Problem Overview, Literature Review, System Architecture, Innovation, and Individual Contributions.
- **Live Code Demonstration:**
 - **Execution:** You must perform a "live run" of your code in the classroom.
 - **Validation:** I will observe the code to verify it meets the technical objectives (e.g., successful model training, data processing, or script execution).
 - **Interactive Q&A:** I will ask technical questions to all members regarding Python principles, library usage, and the implementation logic.